

University of Birmingham
School of Mathematics

Year 2 Tutorials - 2MVA

Semester 1, 2024

Tutorial Sheet 1 - Module Work

These questions are linked to the 2MVA module. These should be attempted in advance of your tutorial meeting in Week 2, unless advised otherwise by your tutor. Solutions will be released after Week 3.

- Q1.** Provide the geometric interpretation of the partial derivative of a function $z = f(x, y)$ to y at the point (a, b) , i.e. $f_y(a, b)$.
- Q2.** Show that $f_{xy} = f_{yx}$ in the following cases
- (a) $f(x, y) = x^m y^n$, where m and n are constants;
 - (b) $f(x, y) = e^{xy} \cos x$.